

# Romain Caneill

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70D5 7116 37B2 9335 9088  
F124 D0FE 114E BFFD ED7F

## Academic experience – employment

- 2024 – 2027 **Postdoc**, *Institute of Environmental Geosciences*, Grenoble, France, working with Pierre Rampal, SASIP project  
Physical oceanography, see ice
- 2018 – 2024 **PhD studies**, *Department of Marine Sciences, University of Gothenburg*, Gothenburg, Sweden, supervised by Fabien Roquet  
Physical oceanography. **From alpha to beta ocean: Exploring the role of surface buoyancy fluxes and seawater thermal expansion in setting the upper ocean stratification**
- 2017 **Master's internship**, *Institute for Geosciences and Environmental research*, Grenoble, France, supervised by Ghislain Picard, 5 months  
**Analysis of Elevation Maps Measured by Laser Scanning at Dome C, Antarctica. Investigation of Snow Accumulation Processes.**
- 2016 **Master's internship**, *Meteorological Institute of Stockholm University*, Stockholm, Sweden, supervised by Fabien Roquet, 3 months  
**Assessing the fine-scale structure of the ocean circulation above the Kerguelen Plateau from instrumented seals.**
- 2015 **Bachelor's internship**, *CEN: Snow Research Centre*, Grenoble, France, supervised by Frédéric Flin, 2 months  
**Calibration of a Cryogenic Cell for the Study of Snow.**

## Education

- 2015 – 2017 **Master's degree in geophysics**, *École Normale Supérieure de Lyon (ENSL) and Lyon 1 University*, Lyon, France  
Physics and chemistry of the Earth and Planets
- 2014 – 2015 **Bachelor's degree in fundamental physics**, *École Normale Supérieure de Lyon and Lyon 1 University*, Lyon, France
- 2012 – 2014 **Two-year university degree in physics and mathematics**, *Paris-Sud University*, Orsay, France

## Teaching experience

- 2022 **CodeRefinery workshop**, *Hosted by coderefinery.org*, 6 half days online – Git version control, testing code, produce reproducible science, Exercise leader, [coderefinery.github.io/2022-03-22-workshop](https://coderefinery.github.io/2022-03-22-workshop)
- 2020 – 2022 **Physical oceanography II (OCM210), Master course**, *Department of Marine Sciences, University of Gothenburg*, Gothenburg, Sweden, Teacher assistant
- 2019 **Ocean modelling (OC6310), Master course**, *Department of Marine Sciences, University of Gothenburg*, Gothenburg, Sweden, Teacher assistant
- 2018 – 2022 **Physical oceanography I (OCM100), Master course**, *Department of Marine Sciences, University of Gothenburg*, Gothenburg, Sweden, Teacher assistant

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## Publications

**R. Caneill**, P. Rampal, M. Bourgoïn, D. Marsan, and J. Weiss, “On the multiscale dynamics of the arctic sea-ice pack,” *Submitted to AGU Advances*, 2026, preprint submitted; License: CC-BY 4.0. [Online]. Available: <https://doi.org/10.1029/2026AV002645>

V. Doriot, F. Roquet, D. Nerini, **R. Caneill**, B. Picard, and C. Guinet, “Distribution and recent trends of the mixed layer depth around the kerguelen plateau,” *Submitted to JGR: ocean*, 2026, preprint submitted; License: Non Exclusive No Reuse License. [Online]. Available: <https://doi.org/10.22541/essoar.173687466.60535825>

**R. Caneill**, F. Roquet, and J. Nycander, “The southern ocean deep mixing band emerges from a competition between winter buoyancy loss and upper stratification strength,” *Ocean Science*, vol. 20, no. 2, pp. 601–619, 2024. [Online]. Available: <https://os.copernicus.org/articles/20/601/2024/>

F. Roquet, D. Ferreira, **R. Caneill**, D. Schlesinger, and G. Madec, “Unique thermal expansion properties of water key to the formation of sea ice on Earth,” *Science Advances*, vol. 8, no. 46, Nov. 2022. [Online]. Available: <https://www.science.org/doi/10.1126/sciadv.abq0793>

**R. Caneill**, F. Roquet, G. Madec, and J. Nycander, “The Polar Transition from Alpha to Beta Regions Set by a Surface Buoyancy Flux Inversion,” *Journal of Physical Oceanography*, vol. 52, no. 8, pp. 1887–1902, Aug. 2022. [Online]. Available: <https://journals.ametsoc.org/view/journals/phoc/52/8/JPO-D-21-0295.1.xml>

G. Picard, L. Arnaud, **R. Caneill**, E. Lefebvre, and M. Lamare, “Observation of the process of snow accumulation on the Antarctic Plateau by time lapse laser scanning,” *The Cryosphere*, vol. 13, no. 7, pp. 1983–1999, 2019. [Online]. Available: <https://tc.copernicus.org/articles/13/1983/2019/>

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## Conferences – workshops

- 2023 **IUGG 2023 Berlin**, *Poster*, Investigating surface buoyancy flux and Ekman transport influence on the Southern Ocean’s upper ocean pycnocline stratification, **R. Caneill**, F. Roquet, G. Madec, J. Nycander  
<https://romaincaneill.fr/news/2023/07/12/iugg23.html>
- 2023 **EGU 23**, *Presentation*, The Influence of Surface Buoyancy Flux and Ekman Transport on Upper Ocean Pycnocline Stratification in the Southern Ocean, **R. Caneill**, F. Roquet, G. Madec, J. Nycander  
<https://doi.org/10.5194/egusphere-egu23-11655>
- 2022 **Workshop in Bornö (Sweden)**, *Title of the workshop: Drivers of the global overturning circulation: wind versus buoyancy*, Organized by Fabien Roquet
- 2021 **virtual EGU 21**, *vPICO*, What determines the position of the transition zone between alpha and beta regions in the ocean? A model study, **R. Caneill**, F. Roquet, G. Madec, J. Nycander  
<https://doi.org/10.5194/egusphere-egu21-14331>
- 2020 **DRAKKAR meeting**, *Poster*, Sensitivity of Oceanic Fronts to Nonlinearities of Equation of State Investigated Using Numerical Experiments, **R. Caneill**, F. Roquet, G. Madec, J. Nycander  
<https://github.com/rcaneill/DRAKKAR-2020-poster/blob/master/poster.pdf>

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## Skills

- Dynamical and descriptive physical oceanography
- Climate and ocean data analysis
- Ocean modelling (run NEMO on HPC, analyze models outputs)

- Profiles analyzes (e.g. ARGO)

### Computer science

- Proficient with **Python** data analysis (numpy, xarray, matplotlib, jupyter, xgcm), and **Snakemake**
- Good level in open- and reproducible-science practices (**Git**, **GitHub**, **GitHub actions**, **GNU/Linux**, **Apptainer** container system, python application testing with **pytest**), **Snakemake**, basics level in **GNU make**
- Proficient with articles and documents production (e.g. **Latex**, **Markdown**, **HTML**)
- Basic level in **C**, **C++**, **Fortran**

### Languages

**French** Native language

**English** Oral and written practice

## Participation to open-source projects

- xnemogcm** Core developer. xnemogcm is a python package that opens NEMO outputs and make them compliant with xgcm.  
[github.com/rcaneill/xnemogcm](https://github.com/rcaneill/xnemogcm)
- xnemogcm-test-data** Containers that can run NEMO 3.6 to 4.2 test case in a reproducible way to produce test data for xnemogcm.  
[github.com/rcaneill/xnemogcm\\_test\\_data](https://github.com/rcaneill/xnemogcm_test_data)
- xgcm** I wrote the NEMO example, the documentation on grid boundary conditions, and I participated to discussions about vertical remapping.  
[github.com/xgcm/xgcm/commits?author=rcaneill](https://github.com/xgcm/xgcm/commits?author=rcaneill)
- gsw-xarray** Core developer. gsw-xarray is a xarray wrapper for gsw that adds CF attributes.  
[github.com/DocOtak/gsw-xarray](https://github.com/DocOtak/gsw-xarray)

## Hobbies

**Carpentry** Professional French Diploma in cabinet making (in French: *CAP de menuiserie*).

**Music** Guitar, composition, concert sound system installation.

**Sports** Mountaineering, climbing in competition, skiing, trekking, wave surfing, kite surfing.